

# School Management System

Dynamic Multi-Client White-Label Architecture, Algorithm & Flowchart

Engine: Google Cloud Firestore (Spark Free Plan) | 500,000+ Students Capacity

## 1. Architecture Overview & Core Concept

This system utilizes a dynamic white-label client-isolated architecture. When distributing to any school, the software dynamically connects to the client's own private Google Firebase instance without modifying source code or exposing developer credentials. The developer pays \$0 in hosting, while client schools retain 100% data ownership and privacy.

## 2. Technical Algorithm (Step-by-Step Logic)

### [ PHASE A: SYSTEM INITIALIZATION & DYNAMIC RESOLUTION ]

1. START Application boot.
2. Query LocalStorage for key 'CUSTOM\_FIREBASE\_CONFIG':
  - IF exists: activeConfig = JSON.parse(CUSTOM\_FIREBASE\_CONFIG)
  - ELSE: activeConfig = DEFAULT\_ENV\_VARIABLES (.env)
3. Execute initializeApp(activeConfig) -> instantiate db & storage.
4. Asynchronously fetch initial collections: students, teachers, app\_data.

### [ PHASE B: AUTO-REGEX PARSER & ZERO-TOUCH ONBOARDING ]

5. Admin opens 'Cloud Database' modal and pastes raw snippet.
6. Regex Parser extracts: apiKey, projectId, storageBucket, appId.
7. Auto-populates all form input fields.
8. On 'Save & Connect': saves config to LocalStorage and reloads app.

### [ PHASE C: RUNTIME CRUD & BATCH OPERATIONS ]

9. StoreContext dispatches mutations via adapter (supabase.from).
10. Adapter executes Firestore writeBatch() in chunks of 450 items.
11. Media/Photos upload directly to client's Firebase Cloud Storage.

## 3. System Flowchart Architecture

### FLOWCHART PIPELINE:

[1] App Boot -> [2] Read LocalStorage (Custom vs Default) -> [3] Init Firebase  
[4] Render UI -> [5] Admin Setup (Paste Snippet) -> [6] Regex Auto-Parser  
[7] Form Auto-Filled -> [8] Save to Storage -> [9] Reload with New DB  
[10] User Mutation -> [11] Adapter writeBatch() -> [12] Client's Private Firestore

## 4. Firebase Free Tier Scaling Limits Matrix

- Database Document Storage: 1.0 GB Free = 500,000+ Students (~2KB/student)
- Media Cloud Storage: 5.0 GB Free = 100,000+ Passport Photos (~50KB/photo)
- Daily Quotas: 50,000 Reads & 20,000 Writes / Day (Sufficient for schools up to 5,000 students)